

Project Reference Number :: AASPL/08/Var./90

Date: 04/08/2026

REQUEST FOR PROPOSAL (RFP)

Development of Cloud-Based Central Monitoring System (CMS)

Issued By:**Ability Automation Solutions Pvt. Ltd.**

1. Introduction

Ability Automation Solutions Pvt. Ltd. invites proposals from experienced software development companies for the design, development, testing, deployment, and support of a **Cloud-Based Central Monitoring System (CMS)** for monitoring a **5,000 MW DC Solar Portfolio**.

The selected software partner shall deliver a complete, production-ready web application, including source code, documentation, deployment support, and warranty.

2. Project Objective

Develop a secure, scalable, cloud-hosted CMS capable of collecting data from multiple client-provided APIs, storing historical data, generating reports, monitoring alarms, and providing web-based dashboards for plant operations and management.

3. Project Scope

The selected software partner shall provide complete software development services including:

A. Web Application Development

- Cloud-Based CMS
- Responsive Web Application
- Login System
- User Management
- Role Based Access
- Dashboard Development

B. Dashboard Development

Develop the following dashboards:

- Portfolio Dashboard
- Plant Dashboard
- Energy Dashboard
- Alarm Dashboard
- Reports Dashboard
- Historical Trend Dashboard
- User Management Dashboard
- System Health Dashboard

C. API Integration

The software shall support

- REST API
- JSON/XML
- API Authentication
- Token Based Authentication
- API Health Monitoring
- Automatic Retry
- Error Logging

D. Data Acquisition

- API Polling Interval **Maximum 1 Minute**
- Configurable Synchronization Time
- Automatic Data Validation
- Missing Data Detection
- Communication Failure Alarm

E. Database

Design a scalable database capable of storing

- Plant Data
- Historical Data
- Alarm Data
- User Data
- Configuration Data
- Audit Logs

4. Historical Data

The application shall

- Store historical data for **minimum 10 Years**
- Support fast data retrieval
- Support trend analysis
- Support historical comparison
- Support data export

5. Reports

The software shall generate

- Daily Report
- Weekly Report
- Monthly Report
- Yearly Report
- Plant Performance Report
- Alarm Report
- Availability Report
- User Activity Report
- Excel Export
- PDF Export

6. Alarm Management

- Alarm Generation
- Alarm Acknowledgement
- Alarm History
- Email Notification
- WhatsApp Notification
- SMS Ready Architecture

7. User Management

Minimum Roles

- Super Admin
- Administrator
- Operator
- Management
- Read Only User

8. Cloud Infrastructure

Software shall support

- AWS
- Microsoft Azure
- Google Cloud

Features

- SSL Security
- Database Backup
- Disaster Recovery
- Auto Scaling
- Monitoring
- Logging

9. Performance Requirements

Description	Requirement
API Polling	Maximum 1 Minute
Dashboard Refresh	Maximum 1 Minute
Concurrent Users	Un Limited
Historical Storage	10 Years
Availability	99.9%
Browser Support	Chrome, Edge, Firefox
Mobile Support	Yes

10. Functional Features

The software shall provide

- Portfolio Monitoring
- Plant Monitoring
- Historical Trends
- Reports
- Alarm Management
- User Management
- Audit Trail
- Dashboard Filter
- Search
- Export to Excel
- Export to PDF

11. Security

- HTTPS
- SSL
- JWT Authentication
- Password Encryption
- Audit Trail
- User Activity Logs

12. Deliverables

Vendor shall provide

- Complete Source Code
- Database Script
- Deployment Package
- API Documentation
- Installation Manual
- User Manual
- Administrator Manual
- Technical Documentation
- Database Design Document
- Architecture Document

13. Testing

Vendor shall perform

- Unit Testing
- Integration Testing
- Performance Testing
- Security Testing
- UAT Support

14. Training

Vendor shall provide

- Administrator Training
- User Training
- Technical Handover

15. Warranty

Minimum

- **12 Months Bug Support**
- Unlimited Bug Fixes
- Remote Technical Support

16. Commercial Proposal

Kindly provide

- Development Cost
- UI/UX Cost
- Cloud Deployment Cost
- Third Party License Cost
- AMC Cost
- Warranty Cost
- Additional Feature Cost

17. Project Schedule

Kindly submit

- Development Timeline
- Team Structure
- Resource Plan
- Milestone Plan

Expected Completion:

3–4 Weeks

18. Company Information

Please submit

- Company Profile
- GST Certificate
- PAN
- CIN
- Team Size
- Similar Projects
- Client References
- Technology Expertise

19. Proposal Submission

Please submit

- Technical Proposal
- Commercial Proposal
- Development Timeline
- Architecture Approach
- Technology Stack
- Warranty Details
- Support Plan

Technology Preference

- **Frontend:** React.js / Angular
- **Backend:** ASP.NET Core or Python (FastAPI/Django)
- **Database:** MS SQL Server
- **Cloud:** AWS / Azure
- **Authentication:** JWT / OAuth 2.0
- **API:** RESTful APIs
- **Version Control:** Gi